AI Chatbot for Green Energy Awareness

Abstract

The AI Chatbot for Green Energy Awareness is an intelligent conversational system that educates users about renewable energy sources, energy conservation, and sustainable living. Using Artificial Intelligence (AI) and Natural Language Processing (NLP), the chatbot understands user queries related to solar, wind, hydro, and bioenergy, carbon footprint reduction, government incentives, and energy-efficient practices. It provides structured responses, real-time suggestions, and eco-friendly tips to promote sustainable energy adoption and encourage responsible energy consumption for a greener future.

Methodology

The project was developed using a structured machine learning approach. A dataset containing common green energy–related questions was created and categorized into areas such as solar energy, wind energy, hydropower, bioenergy, energy conservation, and government incentives. Text preprocessing techniques including tokenization and stop-word removal were applied to enhance the performance and accuracy of the Natural Language Processing (NLP) model.

A classification algorithm was trained to identify the intent and category of user queries. Based on the predicted category, the chatbot provides relevant information, renewable energy guidance, and practical energy-saving suggestions. The system was deployed through a web based interface to enable real-time user interaction. Model performance was evaluated using accuracy and precision metrics to ensure reliable and consistent results.

Result

The AI Chatbot for Green Energy Awareness demonstrated effective intent classification and response generation for renewable energy–related queries. The trained NLP model achieved high accuracy in identifying categories such as solar energy, wind energy, energy conservation, and government incentives, and provided relevant guidance and suggestions. Users were able to receive instant information about adopting green energy solutions, helping them make informed and sustainable energy decisions.

1. Classified green energy queries into multiple renewable energy categories.
2. Provided clear and structured explanations about sustainable energy solutions.
3. Improved user awareness regarding energy conservation and carbon footprint reduction.
4. Demonstrated scalability for future enhancements such as voice input, multilingual support, and integration with smart energy monitoring systems.

Overall, the AI Chatbot proves to be a practical AI-driven solution for promoting green energy awareness by simplifying renewable energy concepts through intelligent conversational assistance